Secant Method

Since $(p-\bar{x})^2$ is deleted then unless p is sufficiently close to $\bar{x}$ N-R method may not converge to the root p ($\bar{x}$ be an initial point $p_0=\bar{x}$).

N-R is extremely powerful technique, but has a major difficulty, the need to know the value of $f'(x)$ at each iterative point.

$$\because f'(p_{n-1})=\lim_{x\to p_{n-2}}\frac{f(x)-f(p_{n-1})}{x-p_{n-1}}$$

$$\therefore f'(p_{n-1})\cong\frac{f(p_{n-2})-f(p_{n-1})}{p_{n-2}-p_{n-1}}=\frac{f(p_{n-1})-f(p_{n-2})}{p_{n-1}-p_{n-2}}$$

using this approximation for $f'(p_{n-1})$ in Newton's formula gives

$$p_n=p_{n-1}-\frac{f(p_{n-1})(p_{n-1}-p_{n-2})}{f(p_{n-1})-f(p_{n-2})}$$

The technique using this formula is called the secant method.

Ex (1):- use secant method to find the smallest root of $f(x)=x\ln x-1$ with error $\varepsilon=0.002$.

Solu. $x_0=1, f(x_0)=1\ln 1-1=1$

$x_1=2, f(x_1)=2\ln 2-1=0.3863$

$$x_{n+1}=x_n-\frac{f(x_n)(x_n-x_{n-1})}{f(x_n)-f(x_{n-1})}$$

$$x_2=2-\frac{0.3863(2-1)}{0.3863+1}=2-0.2787=1.7213$$

$$x_3=1.7213-\frac{-0.0652(-0.2787)}{-0.0652-0.3863}=1.7213-\frac{-0.0182}{-0.4515}=1.7616$$

$$x_4 = 1.7616 - \frac{-0.0025(0.0403)}{-0.025 + 0.0652} = 1.7616 - \frac{-0.0001}{0.0627} = 1.7632$$

$$|x_4 - x_3| = 0.0016$$

$\therefore x_4 = 1.7632$ is a root.

Ex(2):- Find the smallest positive roots of the eq. $f(x) = 3x - 1 - \cos x$ with $\varepsilon = 0.003$.

Solu. $3x - 1 - \cos x = 0 \Rightarrow 3x - 1 = \cos x$

$$x_0 = 0,\ f(0) = -2$$
$$x_1 = 1,\ f(1) = 2 - 0.541 = 1.459$$

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = 1 - \frac{1.459(1-0)}{1.459 + 2} = 1 - 0.4218 = 0.5782$$

$$x_3 = 0.5782 - \frac{-0.1028(0.5782 - 1)}{-0.1028 - 1.459} = 0.5782 - 0.0278 = 0.606$$

$$x_4 = 0.606 - \frac{0.0039(0.0278)}{0.0039 + 0.1028} = 0.606 - 0.001 = 0.605$$

$|x_4 - x_3| = 0.001 < 0.003$

$\therefore x_4 = 0.605$ is a root.

H.W

1. $f(x) = \cos x - x^2, \varepsilon = 0.003$

2. $f(x) = \cos x + x^2, \varepsilon = 0.003$

Algorithm (Secant Method)

```
function [p1,err,k,y]=secant(f,p0,p1,delta,epsilon,max1)
clear;clc;
for k=1:max1
   p2=p1-feval(f,p1)*(p1-p0)/(feval(f,p1)-feval(f,p0));
   err=abs(p2-p1);
   relerr=2*err/(abs(p2)+delta);
   p0=p1;
   p1=p2;
   y=feval(f,p1);
   if (err<delta)|(relerr<delta)|(abs(y)<epsilon),break,end
end
```

Algorithm (Newton Raphson)

```
f=@(p0) (p0)^4+(p0)^2+1;
df=@(p0) 4*(p0)^3+2*(p0);
p0=0.5;
delta=0.0005;
epsilon=0.0005;
max1=10;
for k=1:max1
   p1=p0-feval(f,p0)/feval(df,p0);
   err=abs(p1-p0);
   relerr=2*err/(abs(p1)+delta);
   p0=p1;
   y=feval(f,p0);
   if (err<delta)|(relerr<delta)|(abs(y)<epsilon),break,end
end
disp('Newton Raphson Method');
disp(p0)
```